

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing backtested Java capacity-planning logic that sets upper and lower bounds on the EC2 capacity Amazon Redshift reserves for patching to balance availability against cost

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Rebuilt a legacy test-data workflow tool as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation to make editing smoother, which **90%** of 30+ surveyed developers preferred
- Cut AI workflow generation time **65%** to under 15s by splitting a Claude skill into focused sub-skills that call Python search and validation tools instead of searching open-ended
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a Redis-cached category tree to keep lookups within the database's permission and rate limits, which became the Claude skill's main search tool

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Co-built a research tool that scores and aggregates topics to show Amazon Leo's public-policy staff what's popular, which runs in dev on AWS Fargate with GitHub Actions CI/CD and AWS CDK infrastructure-as-code ready for production
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Built an EventBridge-scheduled LLM-as-judge check for a production Amazon Bedrock classifier that rates research documents by importance to trigger fine-tuning only when its grades shift past a threshold

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D to let a vision-language model pick robot grasps for a PhD-led ICLR project, and tuned its prompts to favor grasps that avoid hitting the table

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a 110+ member student product studio that has shipped 5 live products
- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency
- Building RAG meal suggestions for the official UMD app with SentenceBERT vector search in ChromaDB to feed GPT-4o mini real scraped recipes, which internal testers rated satisfactory for **93%** of meals

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine to replay **50.9M** real IEX exchange messages without ever resyncing, which agreed with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Implemented B+ tree and LSM-tree backends for the engine's event log to compare them on identical 2M-record workloads, which showed the B+ tree giving **12.8x** the read throughput and 22x faster range queries for 27% more disk
- Designed the book as std::list FIFOs per price level inside a std::map with an order-ID hash map to make cancels O(1), which sustained **1.26M events/sec** end-to-end over the full replay

Technical Skills

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL, Bash

Cloud & DevOps: AWS (Lambda, Kinesis, Fargate, EC2, API Gateway, EventBridge, CloudWatch), AWS CDK (infrastructure as code), Docker, GitHub Actions (CI/CD)

Backend & Full-Stack: REST APIs, Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML: LLMs (Claude, Bedrock, GPT-4o mini), retrieval-augmented generation (RAG), fine-tuning and evaluation, PyTorch

Practices & Tools: data structures and algorithms, unit testing (Pytest, Jest), profiling (perf, psutil), Linux, Git, Jira, Agile

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)